# Telegrapher's Generative Model via Kac Flows Notes

## Big picture / Outline

Curves in Wasserstain Spaces

(our Motivition is from here!)

Exploding Velocity Field of Diffusion (probability) flow

1-D Kac Flow & Velocity Field

Multidimension Kac Flow & (conditional) Velocity Field

**Important Notations**: (Review before the proofs)

Wasserstein flow

curves in Wasserstein space ✓

Weakly differentiable ✓

Distribution function as the initial function ✓

sifting property of Dirac mass ✓

## 1D Telegraher Eqn

$$
\begin{aligned}
\partial_{tt} u(t,x) + 2a\, \partial_t u(t,x) &= c^2 \partial_{xx} u(t,x), \\
u(0,x) &= f_0(x), \\
\partial_t u(0,x) &= 0.
\end{aligned}
$$

$H^2(\mathbb{R}^d)$: Hilbert space of f have integrable derivatives up to order 2.

$\mathcal{P}_2(\mathbb{R}^d)$: the complete metric space of prob. meas. with finite $2^{nd}$ moments equipped with:

$\int_{\mathbb{R}^d} \|x\|^2 \mu(dx) < \infty$ introduced a metric

$$W_2(\mu,\nu) = \min_{\pi \in \Pi_{(\mu,\nu)}} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x-y\|^2 \, d\pi(x,y)$$

set of all prob. meas. on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals $\mu$ and $\nu$

"transport plans"

$T: \mathbb{R}^d \to \mathbb{R}^d$ measable map. Pushword measure of $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ is $T_\# \mu := \mu \circ T^{-1}$.

$I$ : bounded interval on $\mathbb{R}$

☆ $AC^2(I; \mathcal{P}_2(\mathbb{R}^d))$ space of abs. cont. curves $\mu_t : I \to \mathcal{P}_2(\mathbb{R}^d)$

$$\exists m \in L^2(I): \quad W_2(\mu_s, \mu_t) \le \int_s^t m(r)\,dr, \quad \forall s \le t, \ t \in I.$$

Think of $\mu_t$ as the mass distribution at time $I$.

Recall for real-valued $f: [a,b] \to \mathbb{R}$, abs. cont. means

$$\exists g \in L^1([a,b]) : |f(t) - f(s)| \le \int_s^t |g(r)|\,dr, \ \forall s < t.$$

Weakly Convergence

$$\mu_t \xrightarrow[t \to t_0]{} \mu_{t_0} \iff \int_{\mathbb{R}^d} \varphi(x)\, \mu_t(dx) \longrightarrow \int_{\mathbb{R}^d} \varphi(x)\, \mu_{t_0}(dx), \ \forall \varphi \in C_b^1$$

if $\mu_t \xrightarrow{weak} \mu_{t_0}$, $\quad \mu_t \in AC(I; \mathcal{P}_2(\mathbb{R}^d)) \iff \exists$ Borel meas. vector field

$$v: I \times \mathbb{R}^d \to \mathbb{R}^d \ \text{ s.t. } \ \partial_t \mu_t + \nabla_x \cdot (\mu_t v_t) = 0.$$

in the distribution sense & $\|V_t\|_{L^2(\mathbb{R}^d,\mu_t)} := \left(\int_{\mathbb{R}^d} \|V_t(x)\|^2 \mu_t(dx)\right)^{\frac{1}{2}} \in L^2(I)$
(weak derivative)

i.e. $\int_I \left(\|V_t\|_{L^2(\mathbb{R},\mu_t)}\right)^2 dt < \infty$

# Velocity explosing of Diffusion flows

$$\partial_t P_t = \frac{\sigma^2}{2}\nabla\cdot(P_t \nabla \log P_t) = \frac{\sigma^2}{2}\Delta P_t \;,\; t\in[0,1],\; \lim_{t\downarrow 0} P_t = \delta_0$$ in weak sense

i.e. $\forall \varphi \in C_c^\infty(\underbrace{(0,1) \times \mathbb{R}^d}_{\text{its compact support}})$, we have

$$\int_0^1\int_{\mathbb{R}^d}\left(-\partial_t\varphi(t,x)P_t(x) + \frac{\sigma^2}{2}\nabla_x\varphi(t,x)\cdot\nabla_x P_t(x)\right)dx\,dt + \int_{\mathbb{R}^d}\varphi(0,x)\delta_0(dx) = 0$$

$\Uparrow$ from

$$\int_0^1\int_{\mathbb{R}}\left(\partial_t P_t(x)\,\varphi(t,x)\right)dx\,dt = \int_0^1\int_{\mathbb{R}^d}\frac{\sigma^2}{2}\Delta P_t(x)\,\varphi(t,x)\,dx\,dt$$

$\swarrow$ IBP

$$\left[\int_{\mathbb{R}^d} P_t(x)\varphi(t,x)\,dx\right]_{t=0}^{t=1} - \int_0^1\int_{\mathbb{R}} P_t\,\partial_t\varphi\,dx\,dt$$

At $t=1$, $\varphi(1,x)=0$ since $\mathrm{supp}(\varphi) = (0,1)\times\mathbb{R}$

At $t\downarrow 0$, $P_t \to \delta_0$

Analytically, $P_t(x) = (2\pi t)^{-\frac{d}{2}} e^{\left(-\frac{\|x\|^2}{2t}\right)}$, $t>0$

for $\sigma^2=1$, $P_1(x) = 2\pi^{-\frac{d}{2}} e^{\left(-\frac{\|x\|^2}{2}\right)} = \underbrace{N(0, I_d)}_{\substack{\text{n-dimensional} \\ \text{Gaussian, mean}=0 \\ \text{cov}=I_d}}$ $\quad V_t(x) = \frac{x}{2t}\ (t>0)$

As ☆ $\partial_t P_t = \frac{1}{2}\Delta P_t \iff \partial_t P_t + \nabla\cdot(P_t V_t) = 0$

with $V_t(x) = -\frac{1}{2}\nabla \log P_t(x)$

Proof:

First, show $V_t(x) = -\frac{1}{2}\nabla \log P_t(x)$ solves $\partial_t P_t = \frac{1}{2}\Delta P_t$.

Working backwards,

$$\underline{P_t V_t} = P_t\left(-\frac{1}{2}\nabla \log P_t\right) \overset{\text{identity}}{=} -\frac{1}{2} P_t \frac{\nabla P_t(x)}{P_t(x)} = \underline{-\frac{1}{2}\nabla P_t(x)}$$

$$\Rightarrow \underline{\nabla\cdot[P_t V_t]} = -\frac{1}{2}\nabla\cdot[\nabla P_t] = \underline{-\frac{1}{2}\Delta P_t}$$

$$\Rightarrow \partial_t P_t = -\nabla\cdot[P_t V_t] = \frac{1}{2}\Delta P_t.$$

Then, $\log P_t = -\frac{d}{2}\log(2\pi t) - \frac{\|x\|^2}{2t}$

$$\nabla \log P_t = 0 - \frac{2x}{2t} = \frac{-x}{t}$$

Thus, $V_t(x) = -\frac{1}{2}\nabla \log P_t(x) = \frac{x}{2t}$ $\square$

Now we do time reverse,

$$\tilde{P}_t(x) := P_{1-t}(x) = (2\pi(1-t))^{-\frac{d}{2}} e^{-\frac{\|x\|^2}{2(1-t)}}$$

where $t \in I = [0,1)$, $\tilde{P}_0 = N(0, I_d)$

Then $\tilde{V}_t(x) = -V_{1-t}(x) = \frac{-x}{2(1-t)}$

see [1, Definition 1.1.1; Theorem 8.3.1]. If in addition

$$\int_I \sup_{x\in B} \|v_t(x)\| + \mathrm{Lip}(v_t, B)\, \mathrm{d}t < \infty \quad \text{for all compact } B \subset \mathbb{R}^d, \tag{4}$$

then the ODE

$$\partial_t \varphi(t,x) = v_t(\varphi(t,x)), \quad \varphi(0,x) = x, \tag{5}$$

has a solution $\varphi : I \times \mathbb{R}^d \to \mathbb{R}^d$ and $\mu_t = \varphi(t,\cdot)_\sharp \mu_0$, see [1, Proposition 8.1.8]. Note that often, Ambrosio, et al (2008).

Then $\frac{d}{dt}\tilde{\varphi}_t(x) = \tilde{v}_t(\tilde{\varphi}_t(x)) = -\frac{1}{2(1-t)}\tilde{\varphi}_t(x)$ , $\tilde{\varphi}_0(x) = x$

$$\Rightarrow \frac{d\tilde{\varphi}_t(x)}{\tilde{\varphi}_t} = -\frac{dt}{2(1-t)} \Rightarrow \tilde{\varphi}_t(x) = x\sqrt{1-t}$$

$$\Rightarrow \tilde{v}(\tilde{\varphi}_t(x)) = \frac{-x}{2\sqrt{1-t}}$$

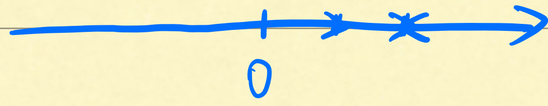
$$\Rightarrow \|\tilde{v}_t\|^2_{L_2(\tilde{p}_t)} = \int_{\mathbb{R}^d} \|\tilde{v}_t(\tilde{\varphi}_t(x))\|^2 d\tilde{p}_0(x) = \frac{d}{4(1-t)}$$

$\forall d \geq 1$

As $\int_0^1 \|\tilde{v}_t\|^2_{L_2(\tilde{p}_t)} dt = \int_0^1 \frac{d}{4(1-t)} dt = \infty$, $\tilde{p}_t \notin AC^2([0,1); P(\mathbb{R}))$.

Infinite Velocity Issue!

# Kac Flow in 1 D



$S_n = c\Delta t \sum_{k=0}^{n} (-1)^{N_k}$ where $N_0 = 0$, $N_k \sim B(k, a\Delta t)$, $k \in \mathbb{N} \setminus \{0\}$.

$= c\Delta t \left(1 + (-1)^{N_1} + (-1)^{N_2} + \cdots (-1)^{N_k}\right)$

$$\boxed{\begin{array}{c}\text{Note as } \Delta t \to 0,\ a\Delta t \to at \text{ as } n \to \infty, \\ N_n \to Poi(at)\end{array}}$$

limit behavior as $n \to \infty$

Let $S(t) = c\tau_t$, $\tau_t := \int_0^t (-1)^{N(s)} ds$

Starts at a random direction then run $S(t)$

Kac Process starting at 0 $K(t) := B_{\frac{1}{2}} S(t) = B_{\frac{1}{2}} c\tau_t$,

Kac Process starting at $x_0$ $:= X_t := X_0 + K(t) = x_0 + B_{\frac{1}{2}} S(t)$

where $B_{\frac{1}{2}} \sim (1, \frac{1}{2})$ a Bernoulli variable taking values in $\pm 1$.

How Kac Flow & Telegrapher Equation connected

weakly differentiable

$v \overset{weak}{=} u'$ if $u, v \in L^1_{loc}(\mathbb{R})$ and $\forall\, \varphi \in C_c^\infty(\mathbb{R})$ smooth compactly supported,

$$\exists v \in L^1_{loc}(\mathbb{R}): \int_{\mathbb{R}} u \varphi'(x) dx = -\int_{\mathbb{R}} v(x) \varphi(x) dx$$

$f: \mathbb{R} \to \ \in L^1_{loc}(\mathbb{R})$ if $\forall K$ compact $\subset \mathbb{R}$, $\int_K |f(x)| dx < \infty$.

$\varphi \in C_c^\infty(\mathbb{R})$ if $\exists [A,B]$ : $\varphi(x) = 0\ \forall x \notin [A,B]$ $\Leftrightarrow$ $supp(\varphi) \subset [A,B]$ is compact

Thm 1 For any initial function $f_0 \in H^2(\mathbb{R})$,

$$u(t,x) = \tfrac{1}{2}E[f_0(x+S(t))] + \tfrac{1}{2}E[f_0(x-S(t))]$$ solves 1D-Telegrapher Equation

Lemma 2 Starting in a generalized (distribution) function, $f_0$ has the distributional solution

$$u(t,x) = \underbrace{\tfrac{1}{2}e^{-at}\big(f_0(x+ct)+f_0(x-ct)}_{①} + \underbrace{\beta ct\int_{x-ct}^{x+ct}\frac{I_0'(\beta r_t(x-y))}{r_t(x-y)}f_0(y)\,dy}_{②}$$
$$\underbrace{+\beta\int_{x-ct}^{x+ct}I_0(\beta r_t(x-y))f_0(y)\,dy\big]}_{③}$$

Example: $f_0 = \delta_0$.

$① = \tfrac{1}{2}e^{-at}(\delta_0(x+ct)+\delta_0(x-ct))$

$\phi(\cdot) \in C_c^\infty(\mathbb{R})$

③: By sifting property of the dirac delta,

$$\int_a^b \phi(y)\delta_0(y)\,dy = \begin{cases}\phi(0), & 0\in[a,b]\\ 0, & 0\notin[a,b]\end{cases}$$

Then $③ = \int_{|x|<ct} I_0(\beta r_t(x-y))f_0(y)\,dy = \beta I_0(\beta r_t(x))\mathbb{1}_{\{|x|<ct\}}$

②: Similarly, by sifting property, $② = \tfrac{1}{2}e^{-at}\beta ct\frac{I_0'(\beta r_t(x))}{r_t(x)}\mathbb{1}_{[-ct,ct]}(x)$

Thus $u(t,x) = \tfrac{1}{2}e^{-at}(\delta_0(x+ct)+\delta_0(x-ct)) + \tilde{u}(t,x)$

where $\tilde{u}(t,x) = ② + ③$. ∎

$\tilde{u}(t,x)$

*This distribution is associated to a probability measure $\mu_t$ with an atomic part given by the Dirac measures $\delta_{-ct}, \delta_{ct}$ and an absolutely continuous part $\tilde{u}(t,\cdot)$, i.e.*

$$\mu_t(A) = \frac{e^{-at}}{2}(\delta_{-ct}(A)+\delta_{ct}(A)) + \int_A \tilde{u}(t,x)\,dx \quad \textit{for all Borel sets } A\subset\mathbb{R}.$$

*The measure $\mu_t$ is exactly the probability distribution of the Kac process $K(t)$ starting in 0 from [15], see [11, Section 2].*

The following remark states an important relation with the left- and right moving Kac processes.

Remark 3 Characterize $\tilde{u}$ w.r.t. $\tilde{u}_+$ & $\tilde{u}_-$

Now Introduce right/left-moving Kac Process starting in 0,

$$\tilde{u}_\pm(t,x) = \frac{1}{2}e^{-at}\Big(\frac{\beta(ct\pm x)}{r_t(x)} I_0'(\beta r_t(x)) + \beta I_0(\beta r_t(x)) \mathbb{1}_{[-ct,ct]}(x)\Big)$$

Note that $\tilde{u} = \frac{1}{2}(\tilde{u}_+ + \tilde{u}_-)$

And corresponding marginal cont. eqn

**Remark 3.** *By [11, Section 2], the absolute continuous part of the distribution of the* initially right-moving *and* initially left-moving *Kac process starting in* 0 *is given by*

$$\tilde{u}_\pm(t,x) = \frac{1}{2}e^{-at}\Big(\frac{\beta(ct\pm x)}{r_t(x)}\, I_0'(\beta r_t(x)) + \beta\, I_0(\beta r_t(x))\Big)1_{[-ct,ct]}(x),$$

*respectively, and it holds* $\tilde{u} = \frac{1}{2}(\tilde{u}_+ + \tilde{u}_-)$ *with* $\tilde{u}$ *in* (19). *By [13, Eq. (12) & (11)], the marginal densities* $\tilde{u}_\pm$ *satisfy the PDEs*

$$\partial_t\tilde{u}_+ + c\,\partial_x\tilde{u}_+ = -a\,\tilde{u}_+ + a\,\tilde{u}_-, \quad \partial_t\tilde{u}_- - c\,\partial_x\tilde{u}_- = a\,\tilde{u}_+ - a\,\tilde{u}_-. \tag{20}$$

*Intuitively, the equations* (20) *can be understood as the continuity equation for the* right-moving *and* left-moving particles *including source and sink terms, see [26]. Adding the equations yields*

$$\partial_t(\tilde{u}_+ + \tilde{u}_-) + c\,\partial_x(\tilde{u}_+ - \tilde{u}_-) = 0.$$

*Thus, with the* continuous flux

$$\tilde{J}(t,x) := \frac{c}{2}\,(\tilde{u}_+(t,x) - \tilde{u}_-(t,x)) = \frac{1}{2}e^{-at}\,\frac{\beta c x}{r_t(x)}\, I_0'(\beta r_t(x)), \quad x \in (-ct, ct). \tag{21}$$

*we arrive at the continuity equation*

$$\partial_t\tilde{u} + \partial_x\tilde{J} = 0. \tag{22}$$

◇

Now we can solve for velocity field of 1D-Kac Process starting in $f_0 = \delta_0$

## Theorem 4

Let $f_0 = \delta_0$. Then $\mu_t$ associated to $u(t,\cdot)$ solves $\partial_t \mu_t = -\partial_x(\mu_t v_t)$ in a weak

& $v_t(x) = v(t,x)$ is given by

$$v(t,x) = \begin{cases} \dfrac{x}{t + \frac{r_t(x)}{c} \frac{I_0 \beta r_t(x)}{I_0'(\beta r_t(x))}} & , \text{ if } x \in (-ct, ct) \\ c & , \ x = ct \\ -c & , \ x = -ct \\ \text{arbitrary} & , \text{ otherwise} \end{cases}$$

Proof: Assume for existance of $v_t$ satisfying $\partial_t \mu_t = -\partial_x(\mu_t v_t)$

Recall $u(t,x) = \underbrace{\frac{1}{2} e^{-at}\left(\delta_0(x+ct) + \delta_0(x-ct)\right)}_{A} + \tilde{u}(t,x)$

Atomic part

$$\partial_t A = \frac{-a}{2} e^{-at}\left(\delta_0(x+ct) + \delta_0(x-ct)\right) + \frac{1}{2} e^{-at}\left(c\left(\delta_0'(x+ct) - \delta_0'(x+ct)\right)\right)$$

$$= -aA(t,x) + \frac{1}{2} e^{-at}\left(c\left(\delta_0'(x+ct) - \delta_0'(x+ct)\right)\right)$$

Recall $\tilde{u}(t,x) := \frac{1}{2}e^{-at}\Big(\underbrace{\beta ct\,\frac{I_0'(\beta r_t(x))}{r_t(x)}}_{A} + \underbrace{\beta I_0(\beta r_t(x))}_{B}\Big)\mathbb{1}_{[-ct,ct]}(x)$

product rule

$$\partial_t \tilde{u}(t,x) = \frac{1}{2}e^{-at}\left(-a(A+B) + \partial_t A + \partial_t B\right)$$

Then Let $C(r) = \beta c\,\frac{I_0'(\beta r)}{r}$

$$\partial_t A = \partial_t\big(t\cdot C(r)\big) = C(r) + t\,C'(r)\,\partial_t r_t$$

where $C'(r) = \beta c\,\frac{d}{dr}\left[\frac{I_0'(\beta r)}{r}\right] \overset{\text{quotient}}{=} \beta c\,\frac{\beta I_0''(\beta r)r - I_0'(\beta r)}{r^2}$

$$\partial_t r_t = \frac{d}{dt}\sqrt{c^2t^2 - x^2} = \frac{c^2t}{\sqrt{c^2t^2-x^2}} = \frac{c^2t}{r_t}$$

Thus, $\partial_t A = \beta c\,\frac{I_0'(\beta r)}{r} + t\beta c\,\frac{\beta I_0''(\beta r)r - I_0'(\beta r)}{r^2}\,\frac{c^2t}{r_t}$

$$\partial_t B = \beta I_0'(\beta r_t)(\beta\,\partial_t r_t) = \beta^2 I_0'(\beta r_t)\frac{c^2t}{r_t}$$

Thus,

$$\partial_t\tilde{u} = \frac{1}{2}e^{-at}\Big\{-a(A+B) + \beta c\,\frac{I_0'(\beta r_t)}{r_t} + t\beta c\,\frac{\beta I_0''(\beta r_t)r_t - I_0'(\beta r_t)}{r_t^2}\,\frac{c^2t}{r_t} + \beta^2 I_0'(\beta r_t)\frac{c^2t}{r_t}\Big)$$

$$\Rightarrow \partial_t u = (a + a^2t)z_1(t,x) + at\,z_2(t,x)$$

where

$$z_1(t,x) := \frac{I_0'(\beta r_t(x))}{r_t(x)} \quad \text{and} \quad z_2(t,x) := \frac{act\,I_0''(\beta r_t(x)) - c^2t\,z_1(t,x)}{r_t(x)^2}.$$

As $\tilde{u}(t,x) = \frac{1}{2}[\tilde{u}_+(t,-x) + \tilde{u}_-(t,-x)]$

$= \frac{1}{2}[\tilde{u}_-(t,x) + \tilde{u}_+(t,x)]$

$= \tilde{u}(t,-x)$

$\Rightarrow \tilde{u}$ symmetric & even $\overset{\text{confinuity eqn}}{\Rightarrow} \partial_x(\tilde{u}v)$ even $\Leftrightarrow \tilde{u}v$ odd in $x$

$\Downarrow$

$v$ odd in $x$ : $v(t,-x) = -v(t,x)$.

From continuity eqn,

$$\int_{-x}^{x} \partial_t \tilde{u}\,dy + \underbrace{\int_{-x}^{x} \partial_y(\tilde{u}v)\,dy}_{①} = 0$$

① $\overset{FTC}{=} u(t,x)\,v(t,x) - \tilde{u}(t,-x)\,v(t,x)$

$\overset{\tilde{u}\text{ even, }v\text{ odd}}{=} 2\tilde{u}(t,x)\,v(t,x)$

$$\Rightarrow 2\tilde{u}(t,x)\,v(t,x) = -\int_{-x}^{x} \partial_t \tilde{u}(t,y)\,dy$$

Similarly, $-\int_{-x}^{x} \partial_t \tilde{u}(t,y)\,dy = 2\tilde{J}(t,x)$ , $\forall x \in (-ct, ct)$

Thus,

$$v(t,x) = \frac{\tilde{J}(t,x)}{\tilde{u}(t,x)} = \frac{e^{-at}\,\frac{\beta cx}{r_t(x)} I_0'(\beta r_t(x))}{e^{-at}\left[\frac{\beta ct}{r_t(x)} I_0'(\beta r_t(x)) + \beta I_0(\beta r_t(x))\right]} = \frac{x}{t + \frac{r_t(x)}{c}\,\frac{I_0(\beta r_t(x))}{I_0'(\beta r_t(x))}},$$

Recall $u(t,x) = \frac{1}{2}e^{-at}\left(\delta_0(x+ct) + \delta_0(x-ct)\right) + \tilde{u}(t,x)$

Then by direct differentiation

Next, we consider the case $x = \pm ct$. For any test function $\phi \in C_c^\infty(\mathbb{R})$, we get

$$\int_{-\infty}^{\infty} \partial_t u(t,x)\phi(x)\,dx = -\frac{ae^{-at}}{2}(\phi(-ct)+\phi(ct)) + \frac{e^{-at}}{2}(-c\phi'(-ct)+c\phi'(ct)) + \int_{-ct}^{ct} \partial_t \tilde{u}(t,x)\phi(x)\,dx.$$

On the other hand, the continuity equation (23) gives

$$\int_{-\infty}^{\infty} \partial_t u(t,x)\phi(x)\,dx = \int_{-\infty}^{\infty} -\partial_x(u(t,x)v(t,x))\phi(x)\,dx \overset{\text{IBP}}{=} \int_{-\infty}^{\infty} u(t,x)v(t,x)\phi'(x)\,dx$$
$$= \frac{e^{-at}}{2}(v(t,-ct)\phi'(-ct)+v(t,ct)\phi'(ct)) + \int_{-ct}^{ct} \tilde{u}(t,x)v(t,x)\phi'(x)\,dx.$$

pull atomic part of $u_t$ out

Now we can choose a sequence of test functions $\phi_n \in C_c^\infty(\mathbb{R})$ such that $\underline{\phi_n \equiv 0 \text{ on } [-ct+\frac{1}{n}, ct-\frac{1}{n}]}$, $\|\phi_n\|_\infty < \frac{1}{n}$, $\phi_n'(ct) = 1$ and $\phi_n'(-ct) = -1$. Then the anti-symmetry $v(t,-ct) = -v(t,ct)$ and above identities in the limit $n \to \infty$ yield

to isolate end pts

$$\frac{e^{-at}}{2}(-c\cdot(-1)+c\cdot 1) = \frac{e^{-at}}{2}(-v(t,ct)\cdot(-1)+v(t,ct)\cdot 1),$$

and hence, $v(t,ct) = c$. By anti-symmetry, it holds $v(t,-ct) = -c$, and the proof of (24) is finished. □

Recall $\partial_t u + \partial_x(uv) = 0$

$$\Rightarrow \int_{-\infty}^{\infty} [\partial_t u + \partial_x uv]\phi\,dx = 0 \underset{\phi \in C_c^\infty}{\overset{\text{IBP}}{\Longrightarrow}} \int_{-\infty}^{\infty} \partial_t u\phi\,dx = \int_{-\infty}^{\infty} uv\phi'\,dx$$

As we can express $\int_{-\infty}^{\infty} \partial_t u\phi\,dx$ in the above 2 forms, we get

$$\frac{e^{-at}}{2}[-c\phi_n'(-ct) + c\phi_n'(ct)] + \underbrace{\int_{-ct}^{ct} \partial_t \tilde{u}\phi_n\,dx}_{A_n \xrightarrow{n\to\infty} 0}$$

$$=$$

$$\frac{e^{-at}}{2}[v(t,-ct)\phi_n'(-ct) + v(t,ct)\phi_n'(ct)] + \underbrace{\int_{-ct}^{ct} \tilde{u}v\phi_n'\,dx}_{B_n \xrightarrow{n\to\infty} 0}$$

As $\lim_{n\to\infty} \|\phi_n\|_\infty \to 0$, $A_n \,\&\, B_n \xrightarrow{n\to\infty} 0$.

Thus, plug in $\phi_n'(ct) = 1$, $\phi_n'(-ct) = -1$, we get $v(t,ct) = c$.

# Multi-dimensional Telegrapher's Equation

## Kac Insertion Method

For the wave equation (undamped)

$$\begin{cases} \partial_{tt} W(t,x) = c^2 \Delta W(t,x) \\ W(0,x) = f_0(x) \\ \partial_t W(0,x) = 0 \end{cases}$$

Insert a stochastic time

As $W$ solves it, let $u(t,x) = E[W_c(\tau_t, x)]$

Then $u$ solves the multi-dimensional telegrapher eqn

$$\begin{cases} \partial_{tt} u(t,x) + 2a\partial_t u(t,x) = c^2 \Delta u(t,x) \\ u(0,x) = f_0(x) \\ \partial_t u(0,x) = 0 \end{cases}$$

where $\tau_t = \int_0^t (-1)^{N(s)} ds$ called Kac Clock

counts the flips

net forward-traveling time

Now aim to show diffusion & telegrapher eqn solns converges.

**Remark 7.** *In 1D, the solution of (27) is given by $w_c(t,x) = \frac{1}{2}(f_0(x+ct) + f_0(x-ct))$, so that (28) reproduces (17). Unfortunately, in higher dimensions, the telegrapher's solution (28) might not stay a probability density anymore. For example, the mass $\int_{\mathbb{R}^3} u(t,x)\,\mathrm{d}x$ in 3D behaves like $1 - e^{-t}$, see [34, Ch. 2.2].* ◇

In the following, we will see that using an analogous insertion method for the diffusion, it is possible to show that the diffusion and telegrapher's solutions converge to each other.

Recall that a Wiener process (Brownian motion) $\mathbf{W} = (\mathbf{W}(t))_{t\geq 0}$, $\mathbf{W}(t) \in \mathbb{R}^d$, $d \geq 1$, is characterized by i) $\mathbf{W}(0) = \mathbf{0}$, ii) independent increments, iii) $\mathbf{W}(t) - \mathbf{W}(s) \sim \mathcal{N}(0, (t-s)I_d)$ for all $0 \leq s < t$, and iv) $\mathbf{W}$ is almost surely continuous in $t$. From now on, we emphasize vector-valued quantities using bold letters.

**Lemma 8.** *For $f_0 \in H^2(\mathbb{R}^d)$, let $w$ be the solution of the undamped wave equation (27) with $c = 1$. Let $W$ be a one-dimensional Wiener process. Then*

$$h(t,x) := \mathbb{E}\left[w\left(\sigma W(t), x\right)\right], \quad \sigma > 0 \tag{30}$$

*solves the heat equation*

$$\partial_t h(t,x) = \frac{\sigma^2}{2}\Delta h(t,x), \qquad (31)$$
$$h(0,x) = f_0(x).$$

The statement can be found in [11], and a proof was given in [8, Theorem 5].

Also, the following Theorem addresses the converges of $h(t,\cdot)$ and $u(t\cdot)$.

**Theorem 9.** *Fix $c_0 > 0$. For $f_0 \in H^2(\mathbb{R}^d)$, let $u$ be given by (28) with $a > 0$, $c \geq c_0$, and $h$ by (30) with $\sigma^2 := \frac{c^2}{a}$. Then there exists a constant $C = C(c_0, d)$ such that*

$$\|h(t,\cdot) - u(t,\cdot)\|_{L_2(\mathbb{R}^d)} \leq C\Big(\frac{1}{at} + \frac{1}{c(ta)^{\frac{1}{2}}}\Big) \quad \textit{for all } t > 0. \qquad (32)$$

*Proof.* It is well-known that the Laplacian $\Delta : H^2(\mathbb{R}^d) \subset L_2(\mathbb{R}^d) \to L_2(\mathbb{R}^d)$ is a self-adjoint linear operator with non-positive spectrum $\sigma(\Delta) = (-\infty, 0]$. By [19, Corollary 2] the solution of (27) with $c = 1$ is given by $w(t,\cdot) = C(t)f_0$, where $(C(t))_{t\geq 0}$ is a uniformly bounded so-called cosine operator function with $\|C(t)\|_{L_2\to L_2} \leq 1$ for all $t \geq 0$. In particular, it follows that $\|w(t,\cdot)\|_{L_2(\mathbb{R}^d)} \leq \|f_0\|_{L_2(\mathbb{R}^d)}$ is uniformly bounded in $t$. Now, [11, Theorem 2] yields the claim. $\square$

**Remark 10.** *By (32), we see that for any $t \geq 0$, the solution $u^{a,c}(t,\cdot)$ of the telegrapher's equation (29) converges to the solution $h(t,\cdot)$ of the diffusion equation (31) for $a, c \to \infty$ with fixed $\sigma^2 = \frac{c^2}{a}$.* $\diamond$

Recall $u(t,x) = E[w_c(\tau_t, x)]$

$h(t,x) = E[w(\sigma W(t), x)]$

# Multi-Dimensional Kac Process

$\vec{K} := (K^1(t), \cdots, K^d(t))$

$\vec{X} := \vec{X}_0 + K(t)$

Recall Probability Distribution Flow (PDF)

Underlying $(\Omega, \mathbb{P})$

For fixed time $t$: $P_{X_t} := (X_t)_{\#}\mathbb{P}$ i.e. $P_{X_t}(A) = \mathbb{P}\{w: X_t(w) \in A\}$

Then $t \to P_{X_t} \in P_2(\mathbb{R}^d)$ is the PDF of the Kac Process

where $P_2(\mathbb{R}^d) = \{\mu \text{ prob. meas. on } \mathbb{R} : \int_{\mathbb{R}^d} \|x\|^2 \mu(dx) < \infty\}$

**Proposition 11.** *Let $P_{\mathbf{X}_0} \in \mathcal{P}_2(\mathbb{R}^d)$. Then, for the probability distribution flow $(P_{\mathbf{X}_t})_{t\geq 0}$ of the d-dimensional Kac process $\mathbf{X}_t$ starting in $\mathbf{X}_0$, the following holds true:*

*i)* Second *$(P_{\mathbf{X}_t})_{t\geq 0}$ is Lipschitz continuous in the Wasserstein space $(\mathcal{P}_2(\mathbb{R}^d), W_2)$ with constant $d^{\frac{1}{2}}c$, and $\mathbf{X}_t$ is almost surely Lipschitz continuous in $\mathbb{R}^d$ with the same constant.* First

*ii) $(P_{\mathbf{X}_t})_{t\geq 0}$ is an absolutely continuous curve in $AC^2([0,T];\mathcal{P}_2(\mathbb{R}^d))$ for any $T > 0$, and admits a velocity field fulfilling*

$$\|v_t\|^2_{L_2(\mu_t)} \leq dc^2 \quad \text{for a.e. } t \in [0,\infty), \tag{34}$$

*iii) The support of $(P_{\mathbf{X}_t})_{t\geq 0}$ grows at most linearly in time.*

Proof:

$$(i)\ \|X_t(w) - X_s(w)\|^2 = \|K(t)(w) - K(s)(w)\|^2$$
$$= \sum_{i=1}^{d} |K_i(t)(w) - K_i(s)(w)|^2$$
$$= \sum_{i=1}^{d} \left|cB_{\frac{1}{2}}(w) \int_s^t (-1)^{N(z)(w)} dz\right|^2$$

As $|B_{\frac{1}{2}}(w)| = 1$, $|(-1)^{N(z)(w)}| = 1$, $\left|cB_{\frac{1}{2}}(w)\int_s^t (-1)^{N(z)(w)} dz\right| \leq c\left|\int_s^t 1\, dz\right| = c(t-s)$

$$= dc^2|t-s|^2$$

$\Rightarrow X_t$ is a.s. Lipshitz cont. in $\mathbb{R}^d$ with constant $d^{\frac{1}{2}}c$. $\square$

Now show $(P_{\vec{X}_t})_{t\geq 0}$ is Lipshitz continuous in $\mathcal{P}_2(\mathbb{R}^d)$ with constant $d^{\frac{1}{2}}c$.

$$\int_{\mathbb{R}^d\times\mathbb{R}^d} \|(x,y)\|^2 \, d\pi(x,y) \overset{\text{definition}}{=} \int_{\Omega\times\Omega} \|(x_0+K(s),\, x_0+K(t))\|^2 \, d\mathbb{P}$$

$$= \int_{\Omega\times\Omega} \|x_0+K(s)\|^2 + \|x_0+K(t)\|^2 \, d\mathbb{P}$$

$$\leq \int_{\Omega\times\Omega} 2\|x_0\|^2 + 2\|K(s)\|^2 + 2\|x_0\|^2 + 2\|K(t)\|^2 \, d\mathbb{P}$$

$$< \infty$$

since $P_{X_0} \in \mathcal{P}_2(\mathbb{R}^d) \Rightarrow E[\|x_0\|^2] < \infty$

$|K_i(u)| = c\left|\int_0^u (-1)^{N^{(i)}(z)} dz\right| \leq cu \Rightarrow \|\vec{K}(u)\| \leq \sqrt{d}\, cu$.

$\Rightarrow \pi$ is admissible transport plan between $P_{X_s}$ & $P_{X_t}$.

Then $W_2^2(P_{\vec{X}_s}, P_{\vec{X}_t}) \underset{\substack{\text{def} \\ \text{of } W^2(\cdot,\cdot)}}{\leq} \int_{\mathbb{R}^d\times\mathbb{R}^d} \|x-y\|^2 \, d\pi(x,y) = \int_{\Omega\times\Omega} \|\vec{X}_s - \vec{X}_t\|^2 \, d\mathbb{P} \leq dc^2|t-s|^2$

Thus, $(P_{X_t})_{t\geq 0}$ is Lip. Cont. in $\mathcal{P}_2(\mathbb{R}^d)$ with constant $d^{\frac{1}{2}}c$. $\square$

(ii) see reference

(iii) Let $\bar{\Omega}_0 := \operatorname{supp}(P_{\vec{X}_0})$. $\bar{\Omega}_t := \bigcup_{x\in\bar{\Omega}_0} B_{d^{\frac{1}{2}}ct}(x)$

As we showed $\|\vec{X}_t(w) - \vec{X}_s(w)\|^2 \leq dc^2|t-s|^2$, $\vec{X}_t \in \bar{\Omega}_t$ $\mathbb{P}$-a.s.

$\Rightarrow \operatorname{supp}(P_{\vec{X}_t}) \subseteq \bar{\Omega}_t$. $\square$

Now we want to generalize Kac Insertion Method into $\vec{K}$.

Recall $u(t,x) = \frac{1}{2}e^{-at}\left(\delta_0(x+ct) + \delta_0(x-ct)\right) + \tilde{u}(t,x)$

**Theorem 12.** *Let $f_0$ be a probability density of a random variable $\mathbf{X}_0 = (X_0^1, ..., X_0^d)$ with independent components such that $f_0(x) = f_0(x^1, \ldots, x^d) = \prod_{i=1}^d f_0^i(x^i)$ and $f_0^i \in H^2(\mathbb{R})$, $i = 1, \ldots, d$. Let $\mathbf{X}(t)$ be a d-dimensional Kac process starting in $\mathbf{X}_0$ with parameters $a > 0$ and $c \geq c_0 > 0$ and with probability density $u_t$. Further, let $\sigma^2 := \frac{c^2}{a}$ and $\mathbf{X}_0 + \sigma\mathbf{W}(t)$ be a d-dimensional Wiener process starting in $\mathbf{X}_0$ with probability density $h_t$. Assume that $\mathbf{X}_0$ is independent of $\mathbf{K}(t)$ and $\sigma\mathbf{W}(t)$, respectively. Then it holds*

$$\|h(t,\cdot) - u(t,\cdot)\|_{L_2(\mathbb{R}^d)} \leq C_d \left(\frac{1}{at^{1+\frac{d-1}{4}}} + \frac{1}{ca^{\frac{1}{2}}t^{\frac{1}{2}+\frac{d-1}{4}}}\right) + R_d(a,c,t),$$

*where the constant $C_d$ depends on $c_0$, and the function $R_d(a,c,t)$ satisfies $\lim_{a,c\uparrow\infty} R_d(a,c,t) = 0$ for any $t$, and $R_d(a,c,\cdot) \in \mathcal{O}(t^{-(\frac{1}{2}+\frac{d}{4})})$ for any $a, c$. Hence, the density of our multi-dimensional Kac process $L_2$-converges to the density of the Wiener process with rate $\mathcal{O}(t^{-(\frac{1}{2}+\frac{d-1}{4})})$.*

*Proof.* First, by assumption, $(X_0^1, ..., X_0^d, K^1(t), ..., K^d(t))$ is (mutually) independent. Hence, $\mathbf{X}_t = (X_0^1 + K^1(t), ..., X_0^d + K^d(t))$ has independent components $X_0^i + K^i(t)$ that are one-dimensional Kac processes starting in $X_0^i \sim f_0^i$. By Theorem 1 and Proposition 5, they admit a probability density $u^i$ solving the 1D telegrapher's equation (29) with initial functions $f_0^i \in H^2(\mathbb{R})$, and hence, $\mathbf{X}_t$ admits the probability density $u(t,x) = \prod_{i=1}^d u^i(t,x^i)$ by the independence of the components.

Analogously, we conclude that $\mathbf{X}_0 + \sigma\mathbf{W}(t)$ has the density $h(t,x) = \prod_{i=1}^d h^i(t,x^i)$, where the marginal densities $h^i$ solve the 1D heat equation (31) with initial functions $f_0^i \in H^2(\mathbb{R})$. Hence, by Theorem 9, the bound (32) holds componentwise with some $C_1 = C_1(c_0) > 0$, i.e.

$$\|h^i(t,\cdot) - u^i(t,\cdot)\|_{L_2(\mathbb{R}^1)} \leq C_1 \left(\frac{1}{at} + \frac{1}{c(ta)^{\frac{1}{2}}}\right) \quad \text{for all} \quad i = 1, ..., d.$$

Now we show the claim by induction on the dimension $d$. The above inequality implies the assertion for $d = 1$. Let the claim be true for $d \geq 1$. Using that for $a_i, b_i \in \mathbb{R}$ it holds

Identity
$$\prod_{i=1}^{d+1} a_i - \prod_{i=1}^{d+1} b_i = (\prod_{i=1}^{d} a_i)(a_{d+1} - b_{d+1}) + b_{d+1}(\prod_{i=1}^{d} a_i - \prod_{i=1}^{d} b_i),$$

it follows by Minkowski's inequality that

$$
\begin{aligned}
\|h(t,\cdot)-u(t,\cdot)\|_{L_2(\mathbb{R}^{d+1})} &= \Big(\int_{\mathbb{R}^{d+1}} |\prod_{i=1}^{d+1} h^i(t,x^i) - \prod_{i=1}^{d+1} u^i(t,x^i)|^2 \,\mathrm{d}x\Big)^{\frac{1}{2}} \\
&\overset{M}{\le} \Big(\int_{\mathbb{R}^{d+1}} |(\prod_{i=1}^{d} h^i(t,x^i))(h^{d+1}(t,x^{d+1}) - u^{d+1}(t,x^{d+1}))|^2 \,\mathrm{d}x\Big)^{\frac{1}{2}} \\
&\quad + \Big(\int_{\mathbb{R}^{d+1}} |u^{d+1}(t,x^{d+1})(\prod_{i=1}^{d} h^i(t,x^i) - \prod_{i=1}^{d} u^i(t,x^i))|^2 \,\mathrm{d}x\Big)^{\frac{1}{2}} \\
&= \Big(\prod_{i=1}^{d} \|h^i(t,x^i)\|_{L_2(\mathbb{R}^1)}\Big) \|h^{d+1}(t,x^{d+1}) - u^{d+1}(t,x^{d+1})\|_{L_2(\mathbb{R}^1)} \\
&\quad + \|u^{d+1}(t,x^{d+1})\|_{L_2(\mathbb{R}^1)} \Big(\int_{\mathbb{R}^d} |(\prod_{i=1}^{d} h^i(t,x^i) - \prod_{i=1}^{d} u^i(t,x^i))|^2 \,\mathrm{d}x\Big)^{\frac{1}{2}}.
\end{aligned}
$$

For the first summand, note that with the one-dimensional heat kernel $K(t,\cdot)$, Young's inequality gives

$$
\|h^i(t,\cdot)\|_{L_2} = \|K(t,\cdot) * f_0^i\|_{L_2} \le \|K(t,\cdot)\|_{L_2} \|f_0^i\|_{L_1} = \|K(t,\cdot)\|_{L_2} = \frac{1}{(8\pi t)^{\frac{1}{4}}}.
$$

Hence, the first summand can be estimated by

$$
\frac{C_1}{(8\pi)^{\frac{d}{4}}} \left( \frac{1}{a t^{1+\frac{d}{4}}} + \frac{1}{c a^{\frac{1}{2}} t^{\frac{1}{2}+\frac{d}{4}}} \right).
$$

By the induction hypothesis, the second summand can be estimated by

$$
\begin{aligned}
&\Big(\|u^{d+1}(t,x^{d+1}) - h^{d+1}(t,x^{d+1})\|_{L_2(\mathbb{R}^1)} + \|h^{d+1}(t,x^{d+1})\|_{L_2(\mathbb{R}^1)}\Big) \\
&\cdot \Big(C_d\Big(\frac{1}{a t^{1+\frac{d-1}{4}}} + \frac{1}{c a^{\frac{1}{2}} t^{\frac{1}{2}+\frac{d-1}{4}}}\Big) + R_d(a,c,t)\Big) \\
\le\ & \Big(C_1\Big(\frac{1}{at} + \frac{1}{c(ta)^{\frac{1}{2}}}\Big) + \frac{1}{(8\pi t)^{\frac{1}{4}}}\Big) \\
&\cdot \Big(C_d\Big(\frac{1}{a t^{1+\frac{d-1}{4}}} + \frac{1}{c a^{\frac{1}{2}} t^{\frac{1}{2}+\frac{d-1}{4}}}\Big) + R_d(a,c,t)\Big).
\end{aligned}
$$

Straightforward multiplication finishes the proof. □

Then we consider the (Conditional Velocity Field) of the Mult-d Wasserstein flow

field $v_t(x|x_0)$ decomposes into univariate velocities $v_t^i(x^i|x_0)$ which are analytically and numerically accessible. These conditional velocity fields can then be used to approximate the true velocity field $v_t$ within the framework of conditional flow matching. But note that conceptually, our velocities are conditioned to start at $t = 0$ in the target measure – reversing the direction of usual flow matching.

The following general lemma states that the velocity field of a multi-dimensional Wasserstein flow can be retrieved by its 1D components if the probability flow admits a multiplicative decomposition (by independence).

**Lemma 13.** *Let $\mu_t \in AC^2(I; \mathcal{P}_2(\mathbb{R}^d))$ such that*

$$\mu_t(x) = \prod_{i=1}^{d} \mu_t^i(x^i) \quad \text{for all } x = (x^1, ..., x^d).$$

*Furthermore, assume that the one-dimensional measure flows $\mu_t^i$ (1D) satisfy a (1D) continuity equation with some vector field $v_t^i$, i.e.*

$$\partial_t \mu_t^i(x^i) = -\partial_{x^i}(\mu_t^i(x^i) v_t^i(x^i)) \quad \text{for all } x^i \in \mathbb{R}.$$

*Then $\mu_t$ satisfies the continuity equation $\partial_t \mu_t(x) = -\nabla \cdot (\mu_t(x) v_t(x))$ with the velocity field*

$$v_t(x) := (v_t^1(x^1), ..., v_t^d(x^d)) \quad \text{for all } x = (x^1, ..., x^d).$$

*Proof.* First, we give a formal derivation of the result, and justify the calculations more rigorously afterwards. We compute $\partial_t \mu_t(x)$ formally by use of the product rule and obtain

$$\partial_t \mu_t(x) = \sum_{i=1}^{d} \Big( \prod_{j \neq i}^{d} \mu_t^j(x^j) \Big) \partial_t \mu_t^i(x^i). \quad \text{product rule}$$

$$\Big\downarrow \ \text{cont. eqn.}$$

Using the continuity equation for $\mu_t^i$, we obtain

$$\partial_t \mu_t(x) = -\sum_{i=1}^{d} \Big( \prod_{j \neq i}^{d} \mu_t^j(x^j) \Big) \partial_{x^i}(\mu_t^i(x^i) v_t^i(x^i)).$$

Pulling the product which does not depend on $x^i$ into the $\partial_{x^i}$-derivative, formally yields the claim

☆ decomposition

$$\partial_t \mu_t(x) = -\sum_{i=1}^{d} \partial_{x^i}(\mu_t(x) v_t^i(x^i)) = -\nabla \cdot (\mu_t(x) v_t(x)).$$

More precisely, we can test above equations by product test functions $\phi(t, x) = \prod_{i=1}^{d} \phi^i(t, x^i)$ with $\phi^i \in C_c^\infty((0, \infty) \times \mathbb{R})$, and obtain the result in the weak formulation. Finally, since the linear subspace generated by such products is dense in $C_c^\infty((0, \infty) \times \mathbb{R}^d)$ and by Lebesgue's dominated convergence, this yields the claim. □

Observe $\mu_t v_t^i(x^i) = \Big(\prod_{j=1}^{d} \mu_t^j(x^j)\Big) v_t^i(x^i) = \mu_t^i(x^i) v_t^i(x^i) \prod_{j \neq i} \mu_t^j(x^j)$

Then $\partial_{x^i}(\mu_t v_t^i(x^i)) = \partial_{x^i}[\mu_t^i v_t^i] \prod_{j \neq i} \mu_t^j$

Thus, $\partial_t \mu_t = -\sum_{i=1}^{d} \partial_{x^i}(\mu_t^i v_t^i) \prod_{j \neq i} \mu_t^j = -\sum_{i=1}^{d} \partial_{x^i}(\mu_t v_t^i) = -\nabla(\mu_t v_t)$

**Theorem 14.** *For a fixed $x_0 \in \mathbb{R}^d$, let $\mathbf{X}_t := x_0 + \mathbf{K}(t)$ be the Kac process starting in $x_0$. Then, $\mu_t(\cdot|x_0) := P_{\mathbf{X}_t}$ satisfies a continuity equation (2) with the velocity field*

$$v_t(x|x_0) := (v_t^1(x^1|x_0^1), \dots, v_t^d(x^d|x_0^d)) \quad \text{for all } x = (x^1, \dots, x^d), \tag{36}$$

*where the univariate components $v_t^i(x^i|x_0^i)$ are given by*

$$v_t^i(x^i|x_0^i) := \begin{cases} (x^i - x_0^i)\left(t + \frac{r_t(x^i - x_0^i)}{c} \cdot \frac{I_0(\beta r_t(x^i - x_0^i))}{I_0'(\beta r_t(x^i - x_0^i))}\right)^{-1} & \text{if } x^i \in (x_0^i - ct, x_0^i + ct), \\ c & \text{if } x^i = x_0^i + ct, \\ -c & \text{if } x^i = x_0^i - ct, \\ \text{arbitrary} & \text{otherwise.} \end{cases} \tag{37}$$

*Furthermore, it holds*

$$v_t^i(x^i|x_0^i) = \mathbb{E}[\dot{X}^i(t) \mid X_t^i = x^i, X_0^i = x_0^i]. \tag{38}$$

by Wald & Steidl

*Proof.* By Proposition 11, $\mu_t(\cdot|x_0)$ admits a velocity field $v_t(\cdot|x_0)$ satisfying the continuity equation (2). By construction, the components $(X_t^1, \dots, X_t^d)$ of $\mathbf{X}_t$ are mutually independent. Hence, $\mu_t(x|x_0) = \prod_{i=1}^d \mu_t^i(x^i|x_0^i)$, where the one-dimensional probability flows $\mu_t^i(\cdot|x_0^i)$ fulfill a continuity equation with 1D velocity fields $v_t^i(\cdot|x_0^i)$. By Lemma 13, it holds the decomposition (36), and Theorem 4 immediately gives the formula (37). Finally, [17, Theorem 3.3], resp. [36, Theorem 6.3] provides (38). $\square$

In reality, we don't have access to $v(t,x)$.

Instead, we define the following loss function based on the conditional velocity fields.

**Corollary 15.** *Let $\mathbf{X}_t$ be the multi-dimensional Kac process starting in $\mathbf{X}_0 \sim \mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$. Assume that $\mathbf{X}_0$ is independent of $\mathbf{K}(t)$, and assume that $\mathbf{X}_0$ has a density $f_0$. Then, the marginal distribution of $\mathbf{X}_t$ admits the density $u_t$ given by*

$$u_t(x) = \int u_t(x|x_0) f_0(x_0)\,\mathrm{d}x_0,$$

*where $u_t(\cdot|x_0)$ is the conditional distribution of $\mathbf{X}_t$ conditioned to $\mathbf{X}_0 = x_0$, and by symmetry*[1]*, $u_t(x|\cdot)$ is exactly the distribution $u_t(\cdot|x)$ for any $x \in \mathbb{R}^d$.*
*Denote the conditional velocity field corresponding to $u_t(\cdot|x_0)$ by $v_t(\cdot|x_0)$. Then, the vector field*

$$v_t(x) := \frac{1}{u_t(x)} \int u_t(x|x_0)\, v_t(x|x_0)\, f_0(x_0)\,\mathrm{d}x_0 \tag{39}$$

*satisfies the continuity equation*

$$\partial_t u_t(x) = -\nabla \cdot \big(u_t(x) v_t(x)\big)$$

*in the distributional sense.*

Now we have all the ingredients to define a feasible loss function for learning the above velocity field $v_t$ by a neural network $v_t^\theta$. Clearly, it is desirable to use the loss

$$\mathcal{L}(\theta) := \mathbb{E}_{(t,x)\sim\mathcal{U}(0,T)\times_t u_t}\left[\left\|v^\theta(t,x) - v(t,x)\right\|^2\right],$$

but we have no direct access to $v$. However, following the same lines as in [15], we see that

$$\mathcal{L}(\theta) = \mathcal{L}_{\mathrm{CFM}}(\theta) + \mathrm{const}$$

with a constant not depending on $\theta$ and the Conditional Flow Matching (CFM) loss is given by

$$\mathcal{L}_{\mathrm{CFM}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}(0,T),\, x_0\sim\mu_0,\, x\sim u_t(\cdot|x_0)}\left[\left\|v^\theta(t,x) - v(t,x|x_0)\right\|^2\right]. \tag{40}$$